

RESEARCH ARTICLE | NOVEMBER 18 2015

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J. Appl. Phys. 118, 194501 (2015)

<https://doi.org/10.1063/1.4935524>



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Equivalent thermal history reconstruction from a partially crystallized glass-ceramic sensor array

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(Received 18 August 2015; accepted 29 October 2015; published online 18 November 2015)

The basic concept of a thermal history sensor is that it records the accumulated exposure to some unknown, typically varying temperature profile for a certain amount of time. Such a sensor is considered to be capable of measuring the duration of several (N) temperature intervals. For this purpose, the sensor deploys multiple (M) sensing elements, each with different temperature sensitivity. At the end of some thermal exposure for a known period of time, the sensor array is read-out and an estimate is made of the set of N durations of the different temperature ranges. A potential implementation of such a sensor was pioneered by Fair *et al.* [Sens. Actuators, A **141**, 245 (2008)], based on glass-ceramic materials with different temperature-dependent crystallization dynamics. In their work, it was demonstrated that an array of sensor elements can be made sensitive to slight differences in temperature history. Further, a forward crystallization model was used to simulate the variations in sensor array response to differences in the temperature history. The current paper focusses on the inverse aspect of temperature history reconstruction from a hypothetical sensor array output. The goal of such a reconstruction is to find an equivalent thermal history that is the closest representation of the true thermal history, i.e., the durations of a set of temperature intervals that result in a set of fractional crystallization values which is closest to the one resulting from the true thermal history. One particular useful simplification in both the sensor model as well as in its practical implementation is the omission of nucleation effects. In that case, least squares models can be used to approximate the sensor response and make reconstruction estimates. Even with this simplification, sensor noise can have a destabilizing effect on possible reconstruction solutions, which is evaluated using simulations. Both regularization and non-negativity constrained least squares minimization approaches are shown to enable robust thermal history estimation. Based on the theoretical and simulation results, several general criteria can be established for the sensor performance and its practical implementation. © 2015 AIP Publishing LLC.
[\[http://dx.doi.org/10.1063/1.4935524\]](http://dx.doi.org/10.1063/1.4935524)

I. INTRODUCTION

For many high temperature engineering problems, accurate estimation of the effect of temperature on component performance is indispensable, since many degradation processes of high temperature materials are thermally activated. The evaluation of such processes often requires knowledge of the accumulated exposure to some varying temperature, i.e., knowledge of the *thermal history*. While thermal history can be measured directly via continuous monitoring using a temperature- or heat flux sensor, sensor data transmission may be cumbersome to implement in some cases. An alternative is provided by a transmission-less *thermal history sensor*, the basic concept of which is that it records *in situ* the accumulated thermal exposure to an arbitrary temperature profile and is read out at the end of a period of use.

Preferably, the sensor readout enables an estimate of a set of exposure times in several different temperature intervals, since in general it is more relevant to know how long some component of interest has been exposed to a high temperature limit than it is to know what the average temperature has been. For this reason, the thermal history sensor

considered here enables measurement of the duration (i.e., time-at-temperature) of several temperature intervals. Thus, although the sensor provides less detail than an exact and entire recording of temperature, it provides more detail than whether some critical value has been exceeded or what the average overall temperature has been. To some extent, a thermal history sensor is similar to the use of a set of classic pyrometric cones which are used during firing of ceramics in order to provide a visual measure of thermal exposure. The deformation of the pyrometric cones is due to a combination of time and temperature. By using several pyrometric cones with different temperature sensitivities and with calibrated deformation rates at various temperatures, such a set of cones can, in principle, be used to estimate the time-at-temperature at several high temperature intervals. The thermal history sensor considered here has the specific objective of reconstructing the duration of several temperature intervals.

Recently, several concepts of thermal history sensors based on crystallization of ceramic materials have been investigated. In the pioneering work by Fair *et al.*, the basic idea was explored of an array of glass-ceramic materials all of which undergo a well-defined devitrification phase transformation, each sensor element having its own unique temperature dependent phase transformation rate.^{1,2} Their simulation

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and experimental data demonstrated that a sensor array enables the distinction between different times-at-temperature events. Most silicate-based glass-ceramics crystallize in the range of 700–1200 °C, which covers most of the temperature range of interest in, for instance, gas turbine hot sections.

The core of the thermal history array sensor is a set of materials that undergo a permanent change in some property, and a range of materials have been investigated recently for this purpose. For instance, several ceramic materials doped with trivalent rare earth ions have been investigated for possible use in a luminescent thermal history sensor,^{3–11} including Tb:Y₂SiO₅,^{3,5} Dy:YSZ/YAG,⁶ Eu²⁺:BaMgAl₁₀O₁₇,^{7,9} Eu²⁺,Mn²⁺:BaMgAl₁₀O₁₇,⁷ Eu²⁺:SrAl₁₄O₂₅,⁷ Dy:YAG and Eu:YAG,⁸ and Eu- and Tb-doped Y₂O₃ and Eu-doped alumina.¹⁰ These materials are also used as thermographic phosphors and are of interest for use in gas turbine combustion environments. Another interesting approach to luminescent thermal history sensing is the use of semiconductor core-shell quantum dot particles,¹² whereby thermally activated diffusion across the core-shell boundary causes the irreversible change in the transition energy. Finally, a sensor material that has been used in gas turbine environments for recording time at temperature is an irradiated SiC crystal undergoing a slow recovery of crystal defects.¹³ Since a single sensor element can provide only one average temperature estimate, and in practice it is often preferable to know how long the sensor has been exposed to different temperatures, the single materials are considered as candidate elements for use in a sensor array. One approach for developing an array of sensor materials is to use one base material and to fine tune the temperature sensitivity by means of variations in concentrations of active and/or co-doped ions. An example of such an approach was presented recently, based on Cr-Zr co-doped Al₂O₃ materials with different Cr and Zr concentrations.¹⁴

While the sensor array concept of Fair *et al.* was analyzed with a forward crystallization model, the inverse reconstruction of a thermal history from some given sensor readout is a non-trivial problem and forms the topic of this paper. The main challenge is that multiple temperature trajectories can result in one particular fractional crystallization of the sensor array, and hence there may exist a vast range of possible solutions. Furthermore, when nucleation is present, the order of time-at-temperature events can also have an impact on the evolution of the sensor array. On the other hand, the exact temperature trajectory is typically not the main objective, but rather some *equivalent thermal history* that is the closest representation of the true thermal history, i.e., the durations of a set of temperature intervals that result in a set of fractional crystallization values—or some other thermally activated process related to a component of interest—which is closest to the one resulting from the true thermal history. The measurement problem, then, is that of finding the best set of N exposure intervals with durations $t_1, t_2, \dots, t_N$ of a predefined set of temperature intervals centered at $T_1, T_2, \dots, T_N$, as shown graphically in Figure 1.

One of the basic questions in the development of a thermal history sensor array considered here, is what method to use for reconstruction of the equivalent thermal history. Moreover, since the equivalent thermal history is an

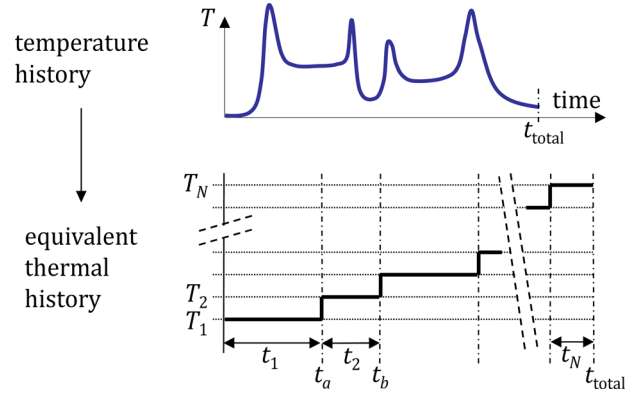


FIG. 1. Conversion of exact temperature history to an equivalent thermal history.

approximation of the true temperature history, the question arises as to what the basic requirements are for an *accurate* reconstruction. And how sensitive the reconstruction is to uncertainties related to sensor measurement and growth parameters. Regarding the reconstruction method, with detailed knowledge of the crystallization dynamics of the sensor elements, the equivalent thermal history can, in principle, be found by using an optimization routine that searches the solution space by means of matching forward crystallization model simulations with sensor output data. However, it can be shown that the search space quickly becomes extremely large. Therefore, a preferred method appears to be to estimate the equivalent thermal history by means of a relatively quick least squares optimization method.

The present paper addresses these questions and forms a first attempt at formulating a method for accurate thermal history reconstruction. While the practical sensor implementation relies on suitable materials with different crystallization rates, the reverse problem of thermal history reconstruction may pose some additional specific requirements on the materials. For instance, it can be shown that thermal history reconstruction is highly complicated by nucleation, and therefore that if nucleation can be neglected, much would be gained. The omission of nucleation can, for instance, be accomplished with sensor materials that exhibit site saturation, using materials with very high nucleation rates or materials that are pre-nucleated. This simplification forms the basic premise of the present paper.

II. THEORY: FORWARD AND RECONSTRUCTION MODEL

We start the outline with the discrete Kolmogorov-Johnson-Mehl-Avrami (KJMA) expression for the degree of crystallinity after j time steps of length Δt , denoted $x(t_j)$, as described by Yinnon and Uhlmann¹⁵

$$x(t_j) = 1 - \exp\left[-\frac{4\pi}{3}\Delta t^4 \Sigma_j\right], \quad \text{where} \quad (1)$$

$$\Sigma_j = \sum_{i=1}^j \left(\sum_{k=i}^j u_k(t_k) \right)^3 I_i(t_i), \quad (2)$$

and where u and I are the temperature and viscosity dependent nucleation, and growth rates defined as¹⁵

$$I(T) = \frac{K}{\eta(T)} \exp \left[-0.0205B \left(1 - \frac{T}{T_m} \right)^{-2} \left(\frac{T}{T_m} \right)^{-3} \right], \quad (3)$$

and

$$u(T) = \frac{10k_B T}{3\pi\eta(T)a_0^2} \left\{ 1 - \exp \left[-\frac{\Delta H_m(T_m - T)}{RT_m T} \right] \right\}, \quad (4)$$

where k_B is Boltzmann's constant, a_0 is the diameter of the rate limiting species, K is a constant equal to $10^{30} \text{ P cm}^{-3} \text{ s}^{-1}$, ΔH_f is the enthalpy of fusion, T_m is the melting point, B is the activation energy barrier to nucleation, and η is the temperature dependent viscosity

$$\eta(T) = A \exp \left(\frac{Q}{T - \tau} \right). \quad (5)$$

Assuming pre-nucleated two-dimensional growth, the master equation Eq. (1) can be simplified to

$$x(t_j) = 1 - \exp \left[-\frac{4\pi}{3} \Delta^2 \left(\sum_{k=1}^j u_k(t_k) \right)^2 \right], \quad (6)$$

such that the fractional crystallization at the end of the intervals of the equivalent thermal history can be defined in a straightforward manner. For instance, at $t = t_a$, one can write

$$x(t_a) = 1 - \exp \left[-\frac{4\pi}{3} (u(T_1) t_1)^2 \right], \quad (7)$$

and similarly at $t = t_b$, the fractional crystallinity can be written as

$$x(t_b) = 1 - \exp \left[-\frac{4\pi}{3} (u(T_1) t_1 + u(T_2) t_2)^2 \right]. \quad (8)$$

When rewriting $u(T_1) = u_1$, after N temperature intervals, one obtains

$$x(t_{total}) = 1 - \exp \left[-\frac{4\pi}{3} (u_1 t_1 + u_2 t_2 + \dots + u_N t_N)^2 \right]. \quad (9)$$

Employing K sensor elements with each having different growth rates, one obtains the following set of equations for the whole sensor array (i.e., for $k = 1, \dots, K$):

$$x_{nk} = 1 - \exp \left[-\frac{4\pi}{3} (u_{1k} t_1 + u_{2k} t_2 + \dots + u_{Nk} t_N)^2 \right]. \quad (10)$$

Eq. (10) can be re-written as a set of K linear equations

$$X_k = \sqrt{-\frac{3}{4\pi} \log(1 - x_{nk})} = u_{1k} t_1 + u_{2k} t_2 + \dots + u_{Nk} t_N, \quad (11)$$

or, defined in matrix form

$$\mathbf{x} = \mathbf{U} \mathbf{t}, \quad (12)$$

where $\mathbf{x}$ is a column vector of length K containing the experimental data from the sensor array, $\mathbf{U}$ is a $K \times N$ matrix, and $\mathbf{t}$ is the column vector containing the durations of the N temperature intervals

$$\mathbf{x} = \begin{pmatrix} \sqrt{-\frac{3}{4\pi} \log(1 - x_{n1})} \\ \vdots \\ \sqrt{-\frac{3}{4\pi} \log(1 - x_{nK})} \end{pmatrix}; \quad \mathbf{U} = \begin{pmatrix} u_{11} & u_{21} & \dots & u_{N1} \\ u_{12} & u_{22} & \dots & u_{N2} \\ \dots & \dots & \dots & \dots \\ u_{1K} & u_{2K} & \dots & u_{NK} \end{pmatrix}; \quad \mathbf{t} = \begin{pmatrix} t_a \\ t_b \\ \vdots \\ t_N \end{pmatrix}. \quad (13)$$

The standard linear least-squares (LLS) solution is then

$$\mathbf{t} = (\mathbf{U}^T \mathbf{U})^{-1} \mathbf{U}^T \mathbf{x}, \quad (14)$$

where $\mathbf{U}^T$ is the transpose of $\mathbf{U}$. It is noted that the dimensionality of growth can have any value other than two in order to allow a similar solution, effectively by changing the exponent in the elements of vector $\mathbf{x}$.

Several observations can be made at this point. First, it can be seen that the thermal history model in the case of site saturation outlined here is not sensitive to the sequence of temperature intervals; i.e., the temperature intervals $T_1, T_2, \dots, T_N$ can occur in any order and result in the same fractional crystallinity. This is also the main reason for utilizing pre-nucleated sensor materials, because in the case of ongoing nucleation the reconstruction is overwhelmingly difficult and highly sensitive to errors, since the temporal order of temperature intervals does matter in that case. It is therefore important to note that this is also a limitation of a thermal history sensor which is insensitive to nucleation, since in many real applications nucleation does play a significant role, and hence also the order of temperature events. Such a simplification is the result of a trade-off between level of sensor detail (i.e., resolution) and ability to compute the equivalent thermal history in a robust manner.

However, despite the simplicity of Eq. (14), it contains a well-known potential source of error, namely, the fact that the set of linear equations can be ill-conditioned. In the present case, the conditioning depends on how similarly the crystallization rates of the sensor elements are defined, and is expressed by the condition number of matrix $\mathbf{U}$. In order to deal with these sources of error, one can use the Tikhonov method of regularization,¹⁶⁻¹⁸ which, essentially, employs a small stabilizing parameter α in the inversion procedure via the unit identity matrix ($\mathbf{I}$), so that the least-squares solution becomes

$$\mathbf{t} = (\mathbf{U}^T \mathbf{U} + \alpha \mathbf{I})^{-1} \mathbf{U}^T \mathbf{x}. \quad (15)$$

In Sec. III, simulations with Eq. (15) are discussed and reveal that while this approach yields reasonably good estimates of equivalent thermal history, the method allows solutions

containing negative durations, especially for the low temperature intervals.

Therefore, an improved reconstruction method was developed that includes a non-negative constraint in vector $\mathbf{t}$, i.e., which solves the following Non-negative Least Squares (NNLS) problem:

$$\begin{aligned} &\underset{\mathbf{t}}{\text{minimize}} \quad f(\mathbf{t}) = \frac{1}{2} \|\mathbf{U}\mathbf{t} - \mathbf{x}\|^2 \\ &\text{subject to } \mathbf{t} \geq 0. \end{aligned} \quad (16)$$

The algorithm developed for this purpose implements a projected quasi-Newton (PQN) method developed by Kim *et al.*¹⁹ The algorithm uses a non-diagonal gradient scaling to solve the NNLS problem which enables fast and robust convergence, even in the presence of noise.

To test the model, the simulation is performed as follows. First, the fractional crystallization of a set of sensor elements with well-defined crystallization parameters is computed for some given temperature profile, using Eqs. (1)–(5). This *forward crystallization model* results in a set of

TABLE I. Parameter list of elements used in thermal history sensor simulations.²

Sensor No.	A (P)	Q	τ	B	T_m (K)	ΔH_f (kJ/mol)
1	10^{-10}	39 000	0	105	1411	280
2	10^{-10}	40 000	0	104	1448	360
3	10^{-10}	41 000	0	103	1484	320
4	10^{-10}	42 000	0	102	1520	200
5	10^{-10}	43 000	0	101	1556	280
6	10^{-10}	44 000	0	100	1592	320
7	10^{-10}	45 000	0	95	1629	200
8	10^{-10}	46 000	0	94	1665	280
9	10^{-10}	47 000	0	93	1701	80
10	10^{-10}	48 000	0	92	1737	240
11	10^{-10}	49 000	0	91	1773	200
12	10^{-10}	50 000	0	90	1809	240
13	10^{-11}	39 000	-40	100	1499	320
14	10^{-11}	41 000	-80	100	1539	360
15	10^{-11}	43 000	-120	100	1578	200
16	10^{-11}	45 000	-160	100	1616	280
17	10^{-11}	46 500	-200	100	1636	160
18	10^{-11}	48 000	-240	100	1655	200
19	10^{-11}	51 000	-300	100	1713	120
20	10^{-11}	53 000	-340	100	1752	200
21	10^{-11}	55 000	-380	100	1791	240
22	10^{-11}	57 000	-420	100	1830	160
23	10^{-11}	58 500	-460	100	1849	320
24	10^{-11}	60 000	-500	100	1868	360
25	10^{-11}	59 000	-460	86	1850	280
26	10^{-11}	58 000	-420	86	1850	200
27	10^{-11}	57 000	-380	87	1850	120
28	10^{-11}	56 000	-320	87	1850	80
29	10^{-11}	54 000	-280	88	1850	240
30	10^{-11}	53 000	-240	88	1850	200
31	10^{-11}	52 000	-200	89	1850	160
32	10^{-11}	51 000	-160	89	1850	320
33	10^{-11}	49 500	-120	90	1850	360
34	10^{-11}	48 000	-80	90	1850	280
35	10^{-11}	47 000	-40	91	1850	200
36	10^{-11}	46 000	0	91	1850	320

hypothetical sensor readout values, under the assumption that the sensor readout is arbitrary accurate. Then, some random multiplicative noise is incorporated in the readout to allow for some inaccuracy in the sensor read-out and/or variability in the sensor crystallization dynamics. The resulting noisy sensor readout vector $\mathbf{x}$ is then fed into the reconstruction algorithm to solve Eq. (15) or (16). An additional constraint included in the algorithm is that the total sensor exposure time is known and can hence be fixed. In Secs. III and IV, a range of simulation results are presented and discussed.

III. RESULTS: SENSOR SIMULATION

Simulation tests were conducted to evaluate the accuracy of thermal history reconstruction from a particular sensor array read-out, and their sensitivity to various parameters, including noise related to, for instance, random sensor variations or read-out errors. For this purpose, the simulations use hypothetical sensor element parameters described in detail in Fair *et al.*,² which are reproduced in Table I.

Figure 2 shows a simulated temperature profile that is used to compute the fractional crystallization of a range of hypothetical sensors. The temperature profile is defined by durations of 600, 400, 300, 200, and 100 s at fixed temperature of 700, 800, 900, 1000, and 1100 °C, respectively. In this example, the durations decrease with higher temperature, which reduces the dominance of the high temperature intervals and effectively results in a large range of possible reconstruction solutions, which is further enhanced when adding noise to the simulation. By changing the simulation conditions, the impact on the spread of the solutions provides an indication of the quality of the reconstruction method.

A sensor array of five elements was assumed, using model parameters of sensors numbered 21 through 25 from Table I. Many different subsets of sensor elements can be chosen from Table I; however, some sets work better than others in terms of attaining a robust and representative solution. Several generalizations on the preferred set of sensor parameters are given in Sec. IV. The fractional crystallinity was computed according to Eq. (6), i.e., with nucleation set

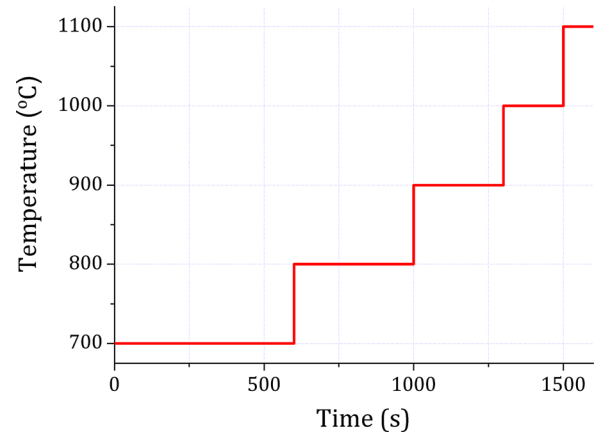


FIG. 2. Simulated temperature profile for testing inverse thermal history model.

to zero and $n=2$. The matrix $\mathbf{U}$ is computed based on the choice of temperature intervals. In the present example, the model was chosen to look for the times at 5 temperature intervals also centered at 700, 800, 900, 1000, and 1100 °C, such that reconstruction can be compared to the input parameters in a straightforward manner for checking accuracy.

As a first test of the algorithm, the errors in the sensor read-out were assumed to be zero; i.e., the computed fractional crystallinity that was used to establish matrix $\mathbf{x}$ was assumed not to have any additional random or systematic errors. Further, the regularization parameter α was set to zero. In this case, the $\mathbf{t}$ vector obtained from the solution of Eq. (15) was $\mathbf{t}^T$ (in seconds) = (600.05 399.94 300.01 200 100), i.e., very close to the simulated input thermal history. Similar results were obtained with a number of other combinations of sensor elements, temperature intervals and times at temperature. It should be noted that even in the error-free cases some combinations produced erroneous results, including large positive and negative values for the elements in the $\mathbf{t}$ vector. This can be attributed to the effect that the matrix problem was unconstrained, i.e., allowing negative elements in $\mathbf{t}$, as well as allowing a total duration that differed from the true duration. Furthermore, some ill-conditioning of the matrix problem exists in several cases, as can be evaluated on the basis of the matrix condition number. The ill-conditioning can be reduced by choosing sufficient sensor elements with differing parameter values.

Next, more realistic scenarios were evaluated whereby the sensor read-out was assumed to have some random noise. For instance, Figure 3 shows the histograms of a large set of simulations assuming 5% random multiplicative noise in the sensor read-out, for three of the five temperature intervals centered at 1100 °C (left), 1000 °C (middle), and 900 °C (right), respectively, and using a stabilization parameter of $\alpha=10^{-11}$, for the input thermal profile of Figure 2. The horizontal axes of the histograms are in seconds and correspond to the estimated time-at-temperature for the particular temperature interval, while the vertical

axis corresponds to the frequency of the estimates in the random number simulations.

The histogram of the temperature interval centered at 1100 °C shows that the mean estimated time-at-temperature (μ) is about 93 s, which is therefore close to the simulated input value of 100 s. Further, the standard deviation (σ) of time-at-temperature estimates is on the order of 7%, which is therefore comparable to the simulated random error in sensor readout. For the second highest temperature interval centered at 1000 °C (middle), the mean value is around 260 s, and therefore larger than the input value of 200 s. Furthermore, the variance in estimates for this temperature interval is significantly larger. Finally, for the third temperature interval of 900 °C (right) and the 4th and 5th at 800 °C and 700 °C (not shown), the mean values are typically closer to zero and the variance is considerable.

Next, we discuss two types of improvements to the type of results shown above and as discussed in a previous article.¹⁴ First, the total exposure time can be added as a constraint, since this parameter is, in principle, known with reasonable accuracy. Second, results will be discussed using a non-negativity constraint as implemented by the NNLS optimization routine.

The resulting thermal history reconstruction using a constrained regularized least squares method, with otherwise the same input values, is shown in Figure 4. As can be seen, the distributions are slightly narrower than in the unconstrained case of Figure 3. Also, the mean values are closer to the real values than in the unconstrained case.

Next, the NNLS routine estimates are shown in Figure 5. As in the previous cases, the mean estimates are denoted μ and can be seen to be comparable to the mean values of the constrained regularized least squares estimates in Figure 4. Two standard deviations are shown; the standard deviation of the complete sample set, σ_s , and that of a Gaussian fit, σ_f , both of which can be seen to be much smaller than in the case of the constrained regularized least squares estimates in Figure 4.

Based on the variance of the individual t -estimates of the equivalent thermal history reconstruction, the NNLS

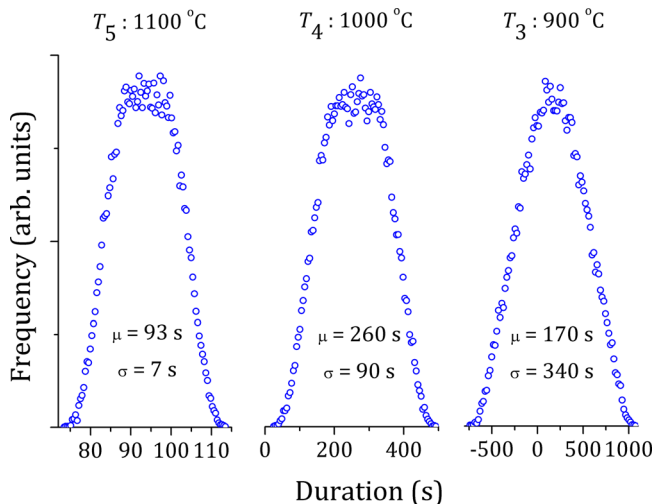


FIG. 3. Histograms of durations at three different temperature intervals, using unconstrained regularized least squares solution (see text for parameter input).

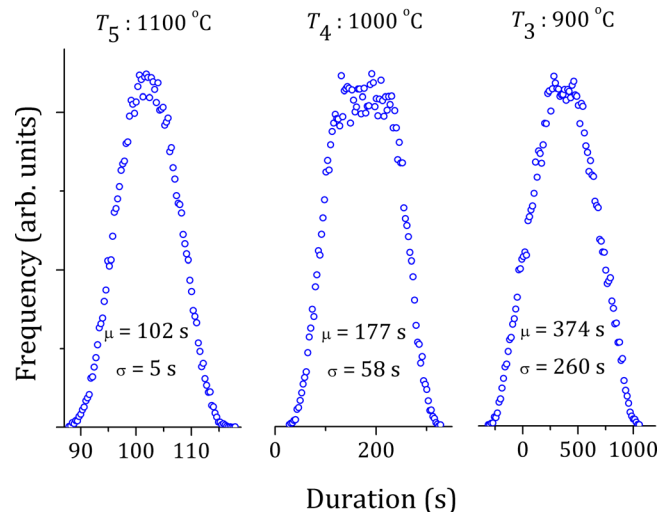


FIG. 4. As in Fig. 3, with total time constrained.

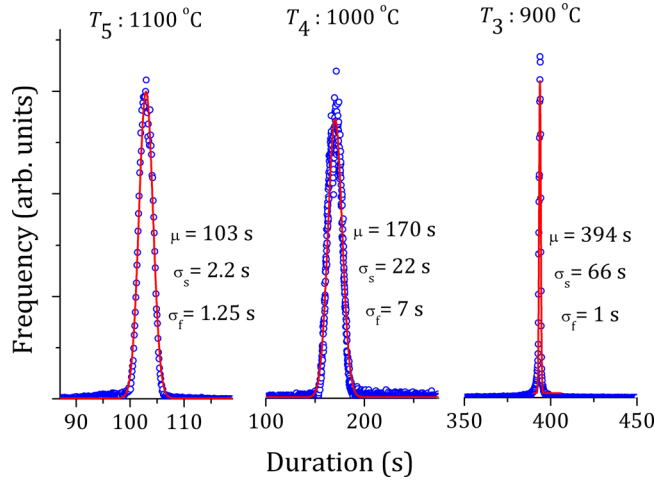


FIG. 5. As in Fig. 3, simulated with PQN-NNLS method.

optimization approach may seem to be preferable. However, what matters is how accurate the *combined set of t-estimates* allows reproduction of the overall thermal process of interest, such as a phase transformation. Since in the case of the regularized LLS method this also depends on the regularization parameter α , the effect of α on the solution will be considered first. A trade-off exists between the trueness and precision of the estimates, depending on the level of noise in the measurements. Therefore, the choice of α depends on the level of uncertainty in the measurement. Simulations were performed to evaluate the sensitivity of the estimates to noise and the regularization parameter. An example is shown in Figure 6, for a noise level of 5%.

As mentioned, even though there is a variance in the estimates of the separate interval durations, the *combination* of estimated durations is what is of main interest. In order to evaluate the accuracy of this set of equivalent thermal history estimates, one can use this set to compute, for instance, the phase transformation of a material of interest. Here, the accuracy is tested by feeding the set of estimates back into the forward crystallization model to compute the end-crystallization of one of the sensor elements. This allows evaluation of the robustness and accuracy of the equivalent thermal history reconstruction. For instance, Figure 7 shows

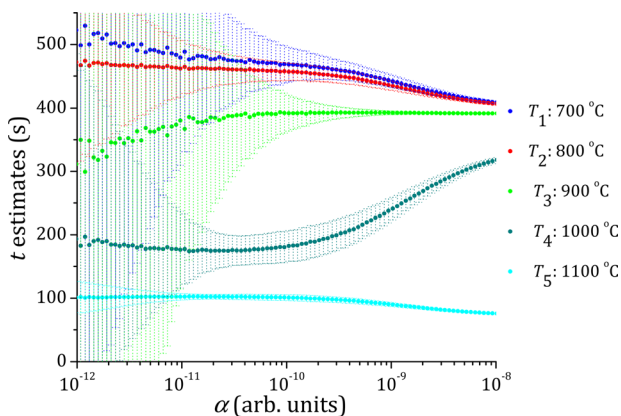


FIG. 6. Estimated interval durations versus regularization parameter, for 5% noise (see text).

the computed end crystallinity of sensor number 21 using the set of t estimates from Figure 6. The exact end crystallinity for sensor 21 upon exposure to temperature history of Figure 2 is $F_{\text{end}} = 0.386$. As can be seen, the regularized LLS method allows accurate estimates of this recomputed fractional crystallinity for values of α larger than approximately 2×10^{-11} . Further, the trueness is within 0.3% and the precision is on the order of 1%, compared to the input random noise of the sensor readout of 5%.

In comparison, the PQN-NNLS optimization routine obtained estimates of F_{end} for sensor #21 with a mean value of 0.385 and a standard deviation of 0.003, i.e., almost the same values as the regularized LLS method with $\alpha > 2 \times 10^{-11}$ and the same input parameters. Thus, the two least squares methods are comparable in accuracy and differ mostly in execution speed and the choice of boundary conditions such as the regularization parameter in the case of LLS.

Finally, it is of interest to briefly discuss the achievable time and temperature resolution of thermal history reconstruction. For instance, it may be of interest to resolve a short temperature spike. As an example, a set of simulations was performed with an input temperature profile as shown in Figure 8, consisting of a total duration of 1000 s at 900°C , 590 s at 1000°C , and a short spike of 10 s, with a temperature (T_{spike}) that was varied for each simulation. A fixed input noise level of 5% was used, and the reconstruction method was based on constrained regularized LLS with $\alpha = 10^{-11}$, using three sensor elements labelled 1–3 from Table I. The reconstruction algorithm used three fixed temperatures T_1 – T_3 of 900, 1000, and 1100°C , respectively. In the case of a spike of 10 s with $T_{\text{spike}} = 1100^\circ\text{C}$, the following duration estimates were obtained: $t_1 = 999.7 \pm 12$ s (representing mean and standard deviation), $t_2 = 590.4 \pm 15$ s, and $t_3 = 9.9 \pm 3.5$ s. Thus, even a short duration spike can be reconstructed, with an accurate mean duration value and a standard deviation representative of the level of input noise. As noted in Sec. II, the combined set of estimated durations (t_1, t_2, t_3) allows higher accuracy in predicting thermal events.

A related aspect is that of temperature resolution, which can be evaluated by systematically changing the input temperature profile while keeping the reconstruction temperature intervals T_1 – T_3 the same. For instance, a variation in T_{spike}

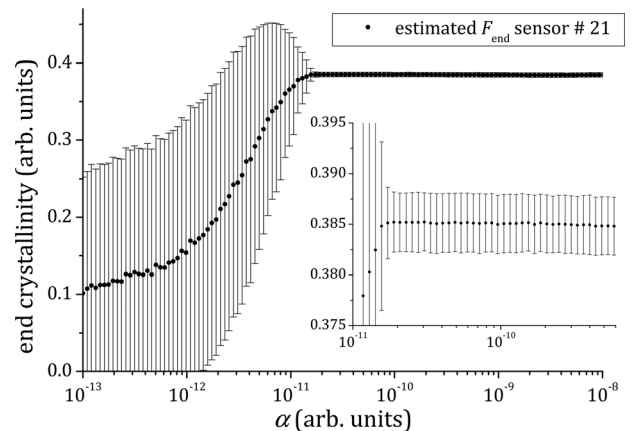


FIG. 7. Example of fractional crystallinity derived from thermal history reconstructions.

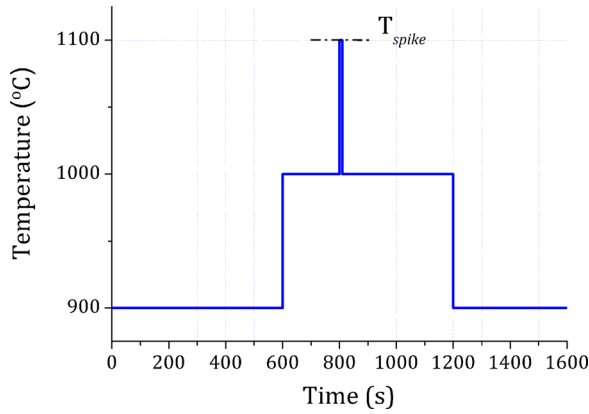


FIG. 8. Simulated temperature profile for testing sensitivity of thermal history reconstruction.

results in a variation of the estimated upper temperature interval duration t_{3_est} as shown in Figure 9. The data point for $T_{spike} = 1100^\circ\text{C}$ was discussed in the previous paragraph, resulting in an accurate mean value $t_{3_est} = 10$ s. For a spike temperature exceeding 1100°C , the effect on the thermal history reconstruction is that the estimated duration t_3 (of the upper temperature interval T_3 centered at 1100°C) increases. This is the consequence of pre-setting the values of the temperature intervals, and from the fact that at higher temperatures the growth rate increases and hence the sensor elements appear to have been exposed for a longer duration at the high temperature interval. The temperature resolution can be roughly approximated by how well the different t_{3_est} values can be distinguished for different T_{spike} values, which in the present example is on the order of $10\text{--}25^\circ\text{C}$, depending on the actual T_{spike} value. However, since this resolution is based on the standard deviation values of only t_3 estimates, this temperature resolution is an underestimation of what appears feasible when taking into account the combined set of estimates.

In any case, it is not so much the separate time or temperature resolution that is of relevance, but rather the combined time-at-temperature resolution. In the present paper, one way to evaluate such lumped parameters, and the accuracy with which they can be estimated from some

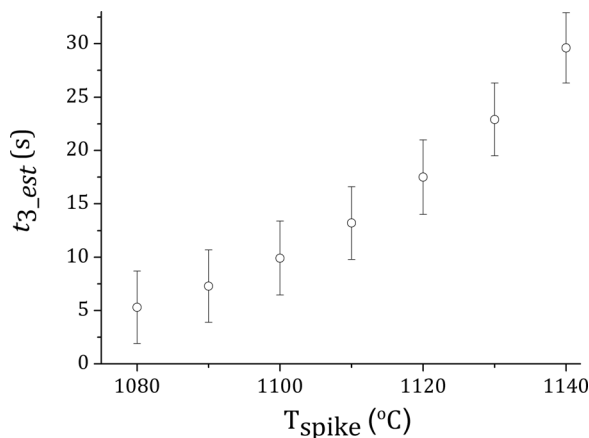


FIG. 9. Simulated high temperature interval duration estimates for several spike temperatures in the temperature profile of Fig. 8.

hypothetical sensor read-out, is to compare the outcome of some thermally activated process with the predictions of the same process based on using the combined set of interval durations from the thermal history sensor.

IV. DISCUSSION AND CONCLUSIONS

In Sec. III, two methods have been described for making accurate estimates of equivalent thermal histories. First, a method based on a constrained regularized linear least squares method allows a robust estimate of durations over several temperature intervals, and subsequent accurate estimation of some thermal exposure process provided that an adequate regularization parameter is chosen. Since this choice depends on the random noise in the measurement, such as encountered by random variations in readout or sensor materials response, some prior knowledge of this uncertainty is needed. The second approach is based on a non-negative least squares optimization routine. An algorithm based on the latter approach may be slightly slower than a direct least squares approach, but nevertheless still be sufficiently fast and robust algorithms such as the one used here (see Ref. 19) have been developed recently that were found suitable for the present purpose. Furthermore, the two approaches were found to yield equivalent thermal history estimates with comparable accuracy.

While the present paper is based on simulations using hypothetical sensor elements with crystallization parameters derived from the work of Fair *et al.*, it provides several considerations for practical implementation.

First, one of the main considerations is related to the role of nucleation. In the present paper, the thermal history sensor theory and simulation was based on a two dimensional growth model without nucleation. If nucleation can be omitted from the sensor model, e.g., in the case of pre-nucleated sensor materials, it can be seen that the sensor model becomes simplified to the point where it can be solved with linear algebra. While the present paper discusses the sensor in terms of a two dimensional growth model, this is not a limitation: the presented model can be shown to be applicable to any dimensional growth case.

A second practical consideration is that the sensor considered here enables reconstruction of multiple temperature intervals, yet in general the high temperature intervals are typically the most important, i.e., carry the most relative weight in the overall results, since the growth rates increase exponentially with temperature. Therefore, the errors in the lower temperature intervals may be of much less concern than errors in the high temperature intervals. Exactly how well the high temperature interval durations need to be estimated will need to be evaluated on a case-by-case basis. In the present case, at least, it was found that although the t -estimates contain errors in the mean values and may contain significant variance, the combined set of t -estimates allows very accurate estimation of a thermally activated process, in this case the fractional crystallization of a another sensor element.

A third consideration is how many sensor elements M are needed for a given equivalent thermal history accuracy

requirement. This, in turn, depends on how the sensors vary in terms of their thermal sensitivity. For instance, for the sensor elements in Table I, sets of parameters can be found that vary sufficiently for the matrix inversion not to be ill-conditioned and that the rank of the sensor matrix is equal to the number of sensor elements. Even if the number of sensor elements is less than the number of temperature intervals of which the duration needs to be estimated, reasonably accurate equivalent thermal history estimates are attainable, depending on which sensor elements are used, and on the temperature input and measurement noise. Therefore, it is of interest to evaluate how different the sensor elements really need to be. Since this question addresses an altogether larger mathematical problem than intended to be addressed in the present work, it suffices to address this issue with several general remarks. The influence of the variation in crystallization parameters on the condition number and rank of matrix $\mathbf{U}$ was found to be approximately as follows. A full rank of matrix $\mathbf{U}$ and lowest condition number were found to be achievable with a relatively small variation in parameters Q and τ of Eq. (5). Next, variations in the parameter ΔH_m over the span shown in Table I were found to yield an almost full rank matrix. The variations in T_m typically resulted in rank 2, whereas the variation in pre-exponential factor A of Eq. (5) yields a rank of 1, i.e., the set of linear equations are linearly dependent when the sensor elements vary only in A and the matrix solution is severely ill-conditioned. Therefore, the best approach for the development of a sensitive and robust thermal history sensor array is to use sensing materials with sufficient variations in viscosity related coefficients Q and τ .

- ¹G. E. Fair, R. J. Kerans, and T. A. Parthasarathy, U.S. patent 6,974,249 (December 13, 2005).
- ²G. E. Fair, R. J. Kerans, and T. A. Parthasarathy, *Sens. Actuators, A* **141**, 245 (2008).
- ³J. P. Feist, J. R. Nicholls, and A. L. Heyes, International Patent WO2009083729 (July 9, 2009).
- ⁴A. Rabhiou, J. P. Feist, A. Kempf, S. Skinner, and A. L. Heyes, in *Proceedings of ASME Turbo Expo 2010: Power for Land, Sea and Air* (2010), Vol. 3, p. 343.
- ⁵A. Rabhiou, J. P. Feist, A. Kempf, S. Skinner, and A. L. Heyes, *Sens. Actuators, A* **169**, 18 (2011).
- ⁶A. L. Heyes, A. Rabhiou, J. P. Feist, and A. Kempf, *AIP Conf. Proc.* **1552**, 891 (2013).
- ⁷A. Rabhiou, A. Kempf, and A. L. Heyes, *Sens. Actuators, A* **177**, 124 (2013).
- ⁸C. C. Pilgrim, A. L. Heyes, and J. P. Feist, *AIP Conf. Proc.* **1581**, 1609 (2014).
- ⁹A. Yáñez González, S. Skinner, F. Beyrau, and A. L. Heyes, in *Proceedings of the International Instrumentation Symposium (ISA)*, London, 2014).
- ¹⁰D. Stenders, I. Karadagli, C. Pflitsch, and B. Atakan, in *Proceedings of the International Instrumentation Symposium (ISA)*, London, 2014).
- ¹¹S. K. Biswas, C. Pilgrim, P. Y. Sollazzo, S. Berthier, and J. P. Feist, in *Proceedings of the International Instrumentation Symposium (ISA)*, London, 2014).
- ¹²Y. Lan, H. Wang, N. Skula, X. Chen, Y. Lu, G. Chen, and Z. Ren, *Part. Part. Syst. Char.* **32**, 65 (2015).
- ¹³J. DeVoe, S. Odom, and G. Ginzburgsky, in *Proceedings of ASME Turbo Expo 2012* (2012), pp. 79–87, Paper No. GT2012-68197.
- ¹⁴B. Heeg, NATO STO scientific publications, MP-AVT-229-07, 2015.
- ¹⁵H. Yinnon and D. R. Uhlmann, *J. Non-Cryst. Solids* **54**, 253 (1983).
- ¹⁶A. N. Tikhonov and V. Y. Arsenin, *Solutions of Ill-Posed Problems* (V. H. Winston and Sons, Washington, DC, 1977).
- ¹⁷C. W. Groetsch, *The Theory of Tikhonov Regularisation for Fredholm Equations of the First Kind* (Pitman Publishing, London, 1984).
- ¹⁸M. Bertero and P. Boccacci, *Introduction to Inverse Problems in Imaging* (Institute of Physics Publishing, Philadelphia, 1998).
- ¹⁹D. Kim, S. Sra, and I. S. Dhillon, “A new projected quasi-Newton approach for the nonnegative least squares problem,” Department of Computer Sciences, The University of Texas at Austin, Technical Report No. TR-06-54, 2006.